

MINECOFIN Rwanda fiscal projection and simulation model

User's manual

June 30, 2021

Overview

Basic features

- The modeling infrastructure is designed to facilitate forecast and scenario creation, as well as model calibration and assessment
- A key feature is the ability to effortlessly create comparison reports
- We can compare different model parameterizations and forecast scenarios
- The results include, among others, equation decompositions, impulse responses, variance decompositions, forecast performance analysis

Software environment

- The infrastructure is implemented in Matlab, using the IRIS macroeconomic modeling toolbox
- Matlab version 2020b and IRIS version 20210329 are required
- The input data is stored in Excel files
- Results are saved in Matlab MAT-files, and also in CSV files (the latter can be opened in Excel as well)
- Reports are generated as PDF files
- It is advisable to use a PDF viewer that does not lock the file (so that the reports can be regenerated while open)
 - A example is Sumatra PDF Reader

Folder structure

- The folder containing all necessary files will be called the *main folder*
- Within the main folder we have the following subfolders:
 - **data** contains the input Excel file
 - **model** contains the model equations and parameter settings
 - **reports** contains the generated PDF reports
 - **results** contains the generated MAT-files and CSV files
- The codes (Matlab M-files) are collected in the package **+codes**
- Besides, there are two files here
 - **mainSettings.m** contains *all* user-defined settings
 - **mainDriver.m** executes the codes in appropriate order

Workflow

- Real life work is usually organized around forecast rounds
- Each forecast round should have its own main folder
- Matlab's current working directory must be the main folder
 - Type `cd` in the Matlab command window to see the current working directory
 - Type `cd(<folder name>)` to set the current working directory
- Adjust `mainSettings.m` according to the task at hand
- Run `mainDriver.m` to execute the necessary commands



Actions

Actions: readData

- `readData.m` reads and processes the data from the Excel file `data/Rwametric.xlsx`. Results are save in `results/data.mat`. There are three databases in this file
 - `dbObs` is the database of the observed variables used in the Kalman filter
 - `dbObsTrans` is the database of the transformed observed variables, e.g. growth rates
 - `dbAux` is the database of the variables necessary to evaluate the equations of the auxiliary model (reporting equations)

Actions: readModel

- `readModel.m` reads the model equations from the model file, sets the parameter values, calculates the steady state and the first order reduced form of the model
- The resulting model object `m` is saved in `results/model.mat`

Actions: filterHistory

- **filterHistory.m** estimates the unobserved variables and shocks of the model using the Kalman filter, optionally with expert judgments (a.k.a. historical tunes)
- Results are saved in **results/filter.mat** There are three databases
 - **dbFilt** the database of filtered variables, for all parameterizations
 - **dbObs** the combined database of observed variables and historical tunes
 - **dbContr** the database of shock contributions, for all parameterizations

Actions: calcForecast

- **calcForecast.m** calculates the forecast with optional expert judgments (a.k.a. forecast tunes)
- If there are multiple parameterizations, the first one will be used
 - It is possible to modify the model parameters in the scenario definition files, but all unmodified parameters will be taken from the first parameterization
- Results are saved in **results/forecast.mat**
 - **dbFcast** is the forecast database, for all scenarios
 - **dbTunes** is a cell array of the tunes for each scenario
 - **m** is the forecast model, for all scenarios
 - **p1n** is a cell array of the simulation plan for each scenario

Actions: calcHistForecast

- `calcHistForecast.m` calculates historical forecasts with the model
- The algorithm loops over each period t in `histForecast.horizon`
- In each iteration
 - It runs the Kalman filter over the range starting in `filterHistory.range(1)` and ending in period $t - 1$ to get *real time estimates* of the unobserved variables
 - It calculates the forecast over a range starting in period t , using the real time estimates of the initial conditions
 - The end of the forecast range is the last period in which all fixed values are available (see the description of settings for further explanation)



Reports

Reports: reportModel

- `reportModel.m` reports the steady states, impulse responses, variance decompositions, implied standard deviations, for multiple parameterizations

Reports: reportHistory

- **reportHistory.m** reports the estimated trends and gaps, historical decompositions, estimated shock series, for multiple parameterizations

Reports: reportForecast

- **reportForecast.m** reports the forecasts of selected variables, decomposition of the equations on the forecast horizon, for multiple sets of forecast tunes and forecast models

Reports: reportHistForecast

- `reportHistForecast.m` plots historical forecast (with an optional list of fixed variables), and shows various forecast performance indicators (RMSE, ME, MAE, RW comparison)

Settings

Settings: basic structure of mainSettings.m

- The file is divided in sections by lines starting with `%%` signs
- The settings are stored in a Matlab structure `opts`
- The settings pertaining to a particular action are collected in fields within `opts`, e.g. `opts.reportHistory`
- Related settings may further be grouped into subfields, e.g. `opts.reportModel.IRF`
- Some time-consuming or unnecessary calculations can be bypassed, e.g. setting `opts.reportHistory.eqDecomps.run = false` will not create the historical decomposition charts

Settings: general options

- **roundId**: round identifier, it will be shown in the report titles and headings
- **modelfile**: name of the file containing the model equations
 - Must not contain the extension, which will automatically be assumed to be **.model**
 - The model file must be in the **model** subfolder of the main folder
 - It is not possible to compare results from different models; however, in many cases it is possible to formulate "model differences" as different parameterizations of the same model

Settings: parameter names

- **parameterNames**: a list of names of parameter files
 - The names must not contain the **.m** extension
 - Each parameter file is a Matlab function, returning the parameter values in a parameter database **p**
 - The parameter files must be in the **model** folder
 - Related parameter files can be collected into subfolders of the **model** folder, in which case the name must contain the subfolder name,
e.g. `["subFolder/setparam1", "subFolder/setparam2"]`
- **parameterLegends**: a list of short, informative descriptions of the parameter settings
 - It will appear in legends, chart title, etc.
 - Should not be very long, otherwise it may cover the chart content

Settings: run settings

- `opts.<action name>.run` determines whether the `mainDriver` will execute that action
- The value can be either `true` or `false`
- For the `reportModel` and `reportHistory` actions, certain subtasks can also be turned-off
- Regardless of the settings here, we can always run an action by executing `code.<action name>` in the command window

Settings: reportModel.steady

- `opts.reportModel.steady` A list of model variables whose steady state values will be included in the report

Settings: reportModel.IRF

- **reportModel.IRF.horizon** Horizon of the impulse response functions. The impulse happens in period 0
- **reportModel.IRF.shocks** List of shocks to include in the report. For each shock there will be a new page in the report showing the responses to the shock.
- **reportModel.IRF.shockSize** Size of the impulse; set it to **@auto** to use one standard deviation shocks
- **reportModel.IRF.variables** List of variables whose responses will be plotted
- **reportModel.IRF.subplot** A numeric vector **[r, c]** of row and column size of the grid of responses. The product **rc** must be at least as large as the number of variables

Settings: reportModel.varDecomp

- **reportModel.varDecomp.horizon** Horizon of the variance decomposition charts
- **reportModel.varDecomp.variables** List of variables whose decomposition will be plotted
- **reportModel.varDecomp.histRange** An IRIS date range over which the historical standard deviations will be calculated
- **reportModel.varDecomp.contribs** An integer **n** indicating how many shocks to name individually
 - The contribution of the **n** most important shocks will be plotted separately, other shock will be grouped together
 - The importance is determined based of the relative contribution on the first period
 - When comparing multiple parameterizations, all shocks which are important in any one parameterization will be plotted separately

Settings: reportModel.stdComp

- **reportModel.stdComp.variables** List of variables to be included in the standard deviations comparison table
- **reportModel.stdComp.shocks** List of shocks to be included in the standard deviations comparison table
- **reportModel.stdComp.histRange** An IRIS date range over which the historical standard deviations will be calculated

Settings: filterHistory

- **filterHistory.range** Range over which the Kalman filter will be run
- **filterHistory.trendGapVars** List of variables whose trend/gap decomposition will be plotted in the report
- **filterHistory.shockDecompVars** List of variables whose shock decomposition will be plotted in the report
- **filterHistory.shockDecompGroups** A structure defining the contents of the shock groups in the shock decomposition charts

Settings: calcForecast

- **forecast.range** Range over which the forecast will be calculated
- **forecast.scenarioNames** List of scenarios for which the forecast will be calculated
 - For each **<scenario>** in this list, the file **<scenario>.m** must exist in the **tunes** subfolder of the main folder
- **forecast.scenarioLegends** A list of short, informative descriptions of the scenario settings
 - It will appear as legends in the forecast report
 - Should not be very long, otherwise it may cover the chart content

Settings: reportForecast

- **reportForecast.plotRange** Range of the forecast plots; besides the forecast range, it may (should) include the recent history as well
- **reportForecast.highlightRange** Range of the forecast plots which will be marked by a different background color; it should either be the forecast range, or the recent history

Settings: calCHistForecast

- **histForecast.range** Range over which the historical forecasts will be calculated
- **histForecast.horizon** The largest horizon for which forecast error statistics will be calculated (in **reportHistForecast.m**)
- **histForecast.variables** List of variables for which the historical forecast plots (a.k.a. "hairy charts") will be created and the forecast error statistics will be calculated
- **histForecast.exogvars** A list of variables/observables/shocks defining the exogenous variables in the historical forecast calculation (see the next slide for details)

Settings: calcHistForecast.exogvars

- It must be a string array with three columns
- In each row
 - the first element is the name of the variable to be exogenized (i.e. fixed)
 - the second element is the name of the corresponding measurement variable
 - the third element is the name of the corresponding shock
- For unobserved variables, the full sample filtered values will be fixed in the Kalman filter
- If the third element is not empty, the observed or the full sample filtered values will be fixed in the forecast



Disclaimer

This document and the information contained within are intended only for the person or entity to whom they are addressed and may contain confidential, proprietary and/or privileged information. Any unauthorized disclosure or distribution to persons or entities other than the intended recipient is prohibited. The information and opinions contained herein do not necessarily reflect OGRResearch's views and OGRResearch does not warrant the correctness thereof. This document is for informational purposes only, it is not intended as a solicitation or offer to enter into any transaction or business relationship, and should not be taken as advice or a recommendation to buy or sell any financial instruments or securities. OGRResearch does not accept liability for any damage caused by reliance on the contents herein.